

How Low-Dimensional Is the S–T–AOU Dependence of Ocean Total Dissolved Inorganic Carbon? A Symbolic Regression Analysis

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September 14, 2026

Abstract

We investigate how low-dimensional the empirical dependence of ocean total dissolved inorganic carbon (TCO_2) on salinity (S), temperature (T), and apparent oxygen utilisation (AOU) is. We analyze 472,530 quality-controlled GLODAP v2.2023 observations across the Atlantic, Indian, Pacific, and Southern Ocean basins. Symbolic regression searches (48 runs across four operator spaces and three random seeds) yielded 599 candidate expressions and 252 Pareto-optimal entries. We evaluate these candidates against a systematic model hierarchy: mean baseline, linear regression, rank-1 quadratic, rank-2 quadratic, full quadratic, and cubic polynomials.

A five-parameter rank-1 quadratic model, $\text{TCO}_2 = (\alpha S + \beta T + \gamma \text{AOU}/100 + \delta)^2 + \epsilon$, captures substantial predictive structure, achieving temporal validation RMSEs of 20.17, 14.48, 16.00, and 10.58 $\mu\text{mol kg}^{-1}$ in the Atlantic, Indian, Pacific, and Southern Ocean, respectively. Crucially, increasing the quadratic rank from one to two provides no meaningful predictive benefit (validation ΔRMSE of +1.98, +0.25, -0.29 , and -0.04 $\mu\text{mol kg}^{-1}$), indicating that a single nonlinear projection accounts for nearly all quadratic curvature. While higher-complexity symbolic expressions achieve 3–12% lower RMSE, they require two- to three-fold higher PySR expression complexity. Rank-1 candidates recurred in 24 of 48 search cells across all four basins,

though recovery was sensitive to operator space—particularly in the Atlantic, where explicit squaring operators were required.

Generalization across spatial-blocked ($5^\circ \times 10^\circ$), cruise-blocked, and locked post-2018 external holdout evaluations demonstrates robust temporal transferability but highlights distinct regional limitations. Antarctic Intermediate Water (AAIW; rank-1 RMSE $28.43 \mu\text{mol kg}^{-1}$, bias $+27.14 \mu\text{mol kg}^{-1}$) and Southern Ocean abyssal waters (global fit $R^2 = -0.174$ vs. local fit $R^2 = 0.786$) represent primary failure regimes. These findings establish that the empirical S – T –AOU relationship is substantially compressible into a compact rank-1 quadratic structure, but its transferability is bounded by domain and water-mass dynamics rather than representing an invariant physical law.

Keywords: total dissolved inorganic carbon; symbolic regression; rank-1 quadratic; GLODAP; model complexity; ocean carbon; low-dimensionality

1 Introduction

Total dissolved inorganic carbon (TCO_2) is a fundamental variable in ocean biogeochemistry, governing air–sea CO_2 flux, ocean acidification, and the biological carbon pump (Sarmiento and Gruber, 2006; Takahashi et al., 1993). Quantifying TCO_2 distributions from hydrographic predictors such as salinity (S), temperature (T), and apparent oxygen utilisation (AOU) is vital for gap-filling sparse observational networks, mapping global carbon inventories, and initializing numerical ocean models (Goyet and Davis, 1991; Key et al., 2004; Broullón et al., 2019).

Traditional statistical formulations range from simple linear regressions (Millero, 1998) to complex multi-layer neural networks (Broullón et al., 2019) and multi-parameter interpolation tools (Carter et al., 2018). While linear models offer parameter efficiency and mathematical simplicity, they struggle to capture the nonlinear curvature driven by carbonate buffer chemistry and water-mass mixing dynamics. Conversely, non-parametric machine learning models attain high predictive precision but act as opaque black boxes, providing limited insight into the intrinsic degrees of freedom governing empirical carbon variability.

Symbolic regression (SR) has emerged as an automated technique for discovering compact closed-form mathematical equations directly from observational datasets (Cranmer, 2023). Rather than treating SR as an automated algorithm for discovering an inviolable "universal law," this study uses symbolic regression as a rigorous probe of empirical low-dimensionality. Specifically, we address the following central question:

How low-dimensional is the empirical S – T –AOU dependence of ocean total dissolved inorganic carbon, and how much predictive accuracy is retained as structural constraints become stronger?

We hypothesize that much of the empirical variability of TCO_2 across the global ocean can be compressed into a low-dimensional nonlinear representation—specifically a five-parameter rank-1 quadratic form—without requiring the full complexity of higher-degree polynomials or deep neural architectures. To test this hypothesis comprehensively, we

evaluate candidate expressions across four major ocean basins (Atlantic, Indian, Pacific, and Southern Ocean) using a suite of 48 symbolic regression searches across four distinct mathematical operator spaces and three random seeds.

Importantly, our work explicitly refrains from claiming:

- (i) the discovery of a universal physical ocean-carbon law;
 - (ii) a first-principles derivation of aqueous carbonate equilibrium;
 - (iii) proof of a single physical mixing axis;
 - (iv) that a rank-1 model is universally optimal or invariant under all validation regimes;
- or
- (v) that symbolic regression uniquely isolated the single "true" equation governing ocean carbon.

Instead, we systematically evaluate the tradeoff between predictive precision and expression complexity across a broad model hierarchy ranging from mean baselines and linear models up to full quadratic polynomials, rank-2 quadratics, cubic regressions, and Pareto-optimal symbolic candidates. We test model transferability using temporal validation, $5^\circ \times 10^\circ$ spatial block cross-validation, cruise-level cross-validation, and a locked post-2018 external holdout dataset. Finally, we map out specific water-mass and abyssal failure regimes, establishing both the utility and the explicit boundaries of compact ocean carbon parameterizations.

2 Data and Partitioning

2.1 GLODAP v2.2023 dataset

Observational data were extracted from the Global Ocean Data Analysis Project version 2 (GLODAP v2.2023) product (Lauvset et al., 2023). GLODAP provides internally consistent, quality-controlled measurements of ocean physical and biogeochemical properties from bottle samples collected on international research cruises. We extracted salinity

(S), in situ temperature (T , °C), apparent oxygen utilisation (AOU, $\mu\text{mol kg}^{-1}$), total dissolved inorganic carbon (TCO_2 , $\mu\text{mol kg}^{-1}$), as well as metadata including latitude, longitude, depth (m), sampling year, GLODAP region code, and cruise identifier.

2.2 Quality control filtering

Strict quality control filters were applied to ensure dataset integrity. Observations were retained only if all primary variables (S , T , AOU, TCO_2) and metadata fields were non-null, finite, and satisfied physically realistic physical boundaries:

$$25.0 < S < 42.0 \text{ PSU}, \quad (1)$$

$$-2.5 < T < 35.0 \text{ }^\circ\text{C}, \quad (2)$$

$$1700.0 < \text{TCO}_2 < 2600.0 \mu\text{mol kg}^{-1}, \quad (3)$$

$$\text{AOU} > -50.0 \mu\text{mol kg}^{-1}. \quad (4)$$

Following these QC filters, a total of 472,530 high-quality global observations were retained for downstream model fitting, validation, and evaluation.

2.3 Basin definitions

To account for distinct oceanographic circulation regimes, ventilation pathways, and water-mass characteristics (Tomczak and Godfrey, 2002), observations were partitioned into four mutually exclusive geographic basins based on latitude and GLODAP region definitions:

- (i) **Atlantic Ocean:** Region code 1, restricted to latitude $\geq -35.0^\circ$ ($N = 147,448$);
- (ii) **Indian Ocean:** Region code 16, restricted to latitude $\geq -35.0^\circ$ ($N = 40,391$);
- (iii) **Pacific Ocean:** Region code 8, restricted to latitude $\geq -35.0^\circ$ ($N = 187,443$); and
- (iv) **Southern Ocean:** All observations with latitude $< -35.0^\circ$ across all longitudes ($N = 97,248$).

2.4 Predictor scaling and notation

In symbolic regression and polynomial modeling, predictor scaling is critical for numerical stability. We scale AOU by a factor of 100 ($\text{AOU}_{100} = \text{AOU}/100$), ensuring that all three predictors (S , T , AOU_{100}) share comparable numerical magnitudes order-of-magnitude wise ($S \sim 35$, $T \sim 0\text{--}30$, $\text{AOU}_{100} \sim -0.5\text{--}4.0$). All candidate models operate directly on these physical variables without standard z-score normalization to preserve direct scientific interpretability.

2.5 Temporal partitioning and external holdout

To evaluate model stability and prevent temporal data leakage, data were partitioned chronologically into three disjoint subsets:

- (i) **Training set** (Years < 2015): $N = 378,507$ observations used exclusively for model coefficient fitting and symbolic expression generation;
- (ii) **Temporal validation set** (Years 2015–2017): $N = 53,474$ observations used for hyperparameter tuning, model family comparison, and Pareto candidate selection; and
- (iii) **Locked external holdout** (Years ≥ 2018): $N = 40,549$ observations strictly reserved for final temporal out-of-sample evaluation.

Importantly, the post-2018 external holdout remained completely locked throughout candidate search and model selection. No hyperparameter adjustment, equation choice, or narrative optimization was performed using post-2018 data. Table 1 summarizes the sample size distributions across basins and temporal splits, and Table 2 details predictor summary statistics.

Table 1: Sample sizes by basin and temporal split.

Basin	N_total	N_train	N_validation	N_external
Atlantic	147448	125762	11436	10250
Indian	40391	32498	3764	4129
Pacific	187443	138202	31791	17450
Southern Ocean	97248	82045	6483	8720
TOTAL	472530	378507	53474	40549

Table 2: Predictor and target distributions by basin.

Basin	Variable	N	Mean	SD	Q25	Median	Q75
Unclassified	salinity	512384	34.665	0.899	34.483	34.694	34.926
Unclassified	temperature	512384	6.707	7.430	1.626	3.562	10.121
Unclassified	aou	512384	101.873	85.448	29.322	85.784	163.690
Unclassified	tco2	512384	2187.727	104.867	2127.000	2182.000	2261.300
Unclassified	depth	512384	1213.613	1380.175	139.000	613.000	1943.925

3 Methods

3.1 Model hierarchy and parameterization

To test the extent of empirical low-dimensionality, we establish a systematic model hierarchy spanning varying degrees of structural constraint and parameter flexibility:

1. **Mean Baseline:** Predicts the basin-specific training mean (1 parameter).

2. **Linear Model:**

$$\text{TCO}_2 = \alpha S + \beta T + \gamma \frac{\text{AOU}}{100} + \delta + \epsilon, \quad (5)$$

fitting 4 parameters with linear combinations of physical state variables.

3. **Rank-1 Quadratic Model:**

$$\text{TCO}_2 = \left(\alpha S + \beta T + \gamma \frac{\text{AOU}}{100} + \delta \right)^2 + \epsilon, \quad (6)$$

fitting 5 coefficients $(\alpha, \beta, \gamma, \delta, \epsilon)$. Mathematically, this enforces an effective rank-1 quadratic Hessian structure $H = 2\mathbf{v}\mathbf{v}^T$, constraining all quadratic curvature to a single one-dimensional direction $\mathbf{v} = (\alpha, \beta, \gamma)^T$.

4. Rank-2 Quadratic Model:

$$\text{TCO}_2 = \left(\alpha_1 S + \beta_1 T + \gamma_1 \frac{\text{AOU}}{100} + \delta_1 \right)^2 + \left(\alpha_2 S + \beta_2 T + \gamma_2 \frac{\text{AOU}}{100} + \delta_2 \right)^2 + \epsilon, \quad (7)$$

fitting 9 parameters $(\alpha_1, \dots, \delta_2, \epsilon)$ to evaluate whether adding a second orthogonal quadratic projection improves predictive capacity.

5. Full Quadratic Model:

$$\begin{aligned} \text{TCO}_2 = & c_0 + c_1 S + c_2 T + c_3 \text{AOU}_{100} + c_4 S^2 + c_5 T^2 + c_6 \text{AOU}_{100}^2 \\ & + c_7 ST + c_8 S \text{AOU}_{100} + c_9 T \text{AOU}_{100} + \epsilon, \end{aligned} \quad (8)$$

fitting 10 parameters, providing the full unconstrained second-order polynomial surface.

6. **Cubic Polynomial Model:** A full third-order polynomial expansion in S, T, AOU_{100} , fitting 20 parameters.

7. **Symbolic Regression Candidates:** Free-form mathematical expressions constructed by PySR, evaluated across varying expression complexities.

We explicitly distinguish between the number of fitted numerical coefficients (K) and PySR expression complexity (C). Expression complexity measure the number of nodes in the symbolic expression tree (e.g., constants, variables, and binary/unary operators). For instance, the rank-1 quadratic model has $K = 5$ fitted parameters and an expression complexity of $C = 8$.

3.2 Symbolic regression implementation

Symbolic regression was conducted using PySR ([Cranmer, 2023](#)) driven by SymbolicRegression.jl in Julia. Searches were performed independently for each basin to discover regional structural relationships. We executed PySR searches across a factorial design of 4 basins $\times$ 4 operator spaces $\times$ 3 random seeds (42, 123, 456), totaling 48 completed

Table 3: Symbolic-regression search configurations.

Search	Binary operators	Unary operators	Max size	Populations	Population size	Iterations
SearchA_basic	+, -, *, /		30	30	50	100
SearchB_explog	+, -, *, /	exp, log	30	30	50	100
SearchC_sqrt_square	+, -, *, /	sqrt, square, exp, log	30	30	50	100
SearchD_broad	+, -, *, /	sqrt, square, exp, log, abs, cube, cbrt	35	40	60	150

evolutionary searches.

Search parameterization maintained strict execution controls:

- Population size: 500 expressions across 33 sub-populations;
- Generations: 100 iterations per run;
- Maximum expression complexity: $C_{\max} = 30$;
- Loss function: Mean Squared Error (MSE); and
- Deterministic single-threaded execution per seed to ensure numerical reproducibility.

3.3 Operator space design

To assess operator dependence, four operator spaces were constructed:

- **Search A (Basic Arithmetic)**: Binary operators $\{+, -, \times, \div\}$;
- **Search B (Exponential/Logarithmic)**: $\{+, -, \times, \div, \exp, \log\}$;
- **Search C (Square/Root)**: $\{+, -, \times, \div, (\cdot)^2, \sqrt{\cdot}\}$; and
- **Search D (Broad Operator Set)**: $\{+, -, \times, \div, (\cdot)^2, \sqrt{\cdot}, \exp, \log\}$.

Table ?? details the exact PySR search configurations.

3.4 Candidate library extraction and Pareto frontier

From the 48 PySR runs, all non-dominated expressions were collected into a unified candidate library of 599 symbolic expressions. For each basin, a global empirical Pareto frontier was constructed by plotting validation RMSE against expression complexity. Candidates were classified into structural equation families (e.g., linear, rank-1 quadratic, rank-2 quadratic, exponential-decay, rational functions) using AST pattern matching.

3.5 Spatial and cruise blocked validation

To test spatial and sampling generalization, two blocked cross-validation schemes were executed on pre-2018 data:

- (i) **Spatial-Block Cross-Validation:** Data were partitioned into spatial tiles of 5° latitude by 10° longitude using GroupKFold cross-validation.
- (ii) **Cruise-Block Cross-Validation:** Data were grouped by GLODAP cruise identifier using GroupKFold cross-validation, evaluating out-of-sample prediction on unobserved research cruises.

3.6 Dependence-aware statistical inference

Because oceanographic bottle samples exhibit spatial and temporal autocorrelation, standard independent error assumptions underestimate standard errors. We performed dependence-aware inference using spatial-block bootstrap resampling (1,000 iterations over $5^\circ \times 10^\circ$ spatial blocks). Paired absolute-error differences ($\Delta\text{AE} = |\text{err}_{\text{rank-1}}| - |\text{err}_{\text{full-quad}}|$) were evaluated across equivalence margins of $\delta = 1.0, 2.0, 5.0, 10.0 \mu\text{mol kg}^{-1}$ using Two One-Sided Tests (TOST) ([Schuirmann, 1987](#)).

3.7 Depth, water-mass, and derivative analysis

Model predictions were stratified across five ocean depth layers (0–100 m, 100–500 m, 500–1000 m, 1000–3000 m, and > 3000 m) and key hydrographic water masses, including Antarctic Intermediate Water (AAIW), North Atlantic Deep Water (NADW), and Antarctic Bottom Water (AABW).

For the rank-1 model, partial derivatives with respect to state variables are:

$$\frac{\partial \text{TCO}_2}{\partial S} = 2\alpha z, \quad \frac{\partial \text{TCO}_2}{\partial T} = 2\beta z, \quad \frac{\partial \text{TCO}_2}{\partial \text{AOU}} = \frac{2\gamma}{100} z, \quad (9)$$

where $z = \alpha S + \beta T + \gamma \text{AOU}_{100} + \delta$. The Hessian matrix of second derivatives is:

$$H = \begin{pmatrix} \frac{\partial^2 \text{TCO}_2}{\partial S^2} & \frac{\partial^2 \text{TCO}_2}{\partial S \partial T} & \frac{\partial^2 \text{TCO}_2}{\partial S \partial \text{AOU}_{100}} \\ \frac{\partial^2 \text{TCO}_2}{\partial T \partial S} & \frac{\partial^2 \text{TCO}_2}{\partial T^2} & \frac{\partial^2 \text{TCO}_2}{\partial T \partial \text{AOU}_{100}} \\ \frac{\partial^2 \text{TCO}_2}{\partial \text{AOU}_{100} \partial S} & \frac{\partial^2 \text{TCO}_2}{\partial \text{AOU}_{100} \partial T} & \frac{\partial^2 \text{TCO}_2}{\partial \text{AOU}_{100}^2} \end{pmatrix} = 2 \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} \begin{pmatrix} \alpha & \beta & \gamma \end{pmatrix} = 2\mathbf{v}\mathbf{v}^T. \quad (10)$$

Because H is formed by an outer product of a single vector $\mathbf{v}$, it has rank exactly 1.

4 Results

4.1 Dataset and baseline reproduction

Quality control filtering of GLODAP v2.2023 yielded 472,530 high-quality global observations across the four ocean basins (Atlantic: 147,448; Indian: 40,391; Pacific: 187,443; Southern Ocean: 97,248). Figure 1 illustrates the global sampling distribution in T – S space colored by TCO_2 . Baseline model fitting confirmed exact reproduction of all primary pipeline results.

4.2 Model-complexity hierarchy

Table 4 and Table 5 compare temporal validation performance across the complete model hierarchy. Linear regression ($K = 4$) achieves validation RMSEs of 21.59, 14.96, 17.77, and $10.66 \mu\text{mol kg}^{-1}$ in the Atlantic, Indian, Pacific, and Southern Ocean, respectively. Introducing a single quadratic direction in the rank-1 model ($K = 5, C = 8$) reduces validation RMSE to 20.17, 14.48, 16.00, and $10.58 \mu\text{mol kg}^{-1}$, delivering immediate performance gains over linear models in all basins.

4.3 Symbolic-regression candidate recurrence

Across the 48 PySR evolutionary searches, a total of 599 candidate expressions were retained, of which 252 entries lay on basin-specific Pareto frontiers. Structurally, 86 expressions belonged to the rank-1 quadratic family. Rank-1 candidates recurred in 24 of

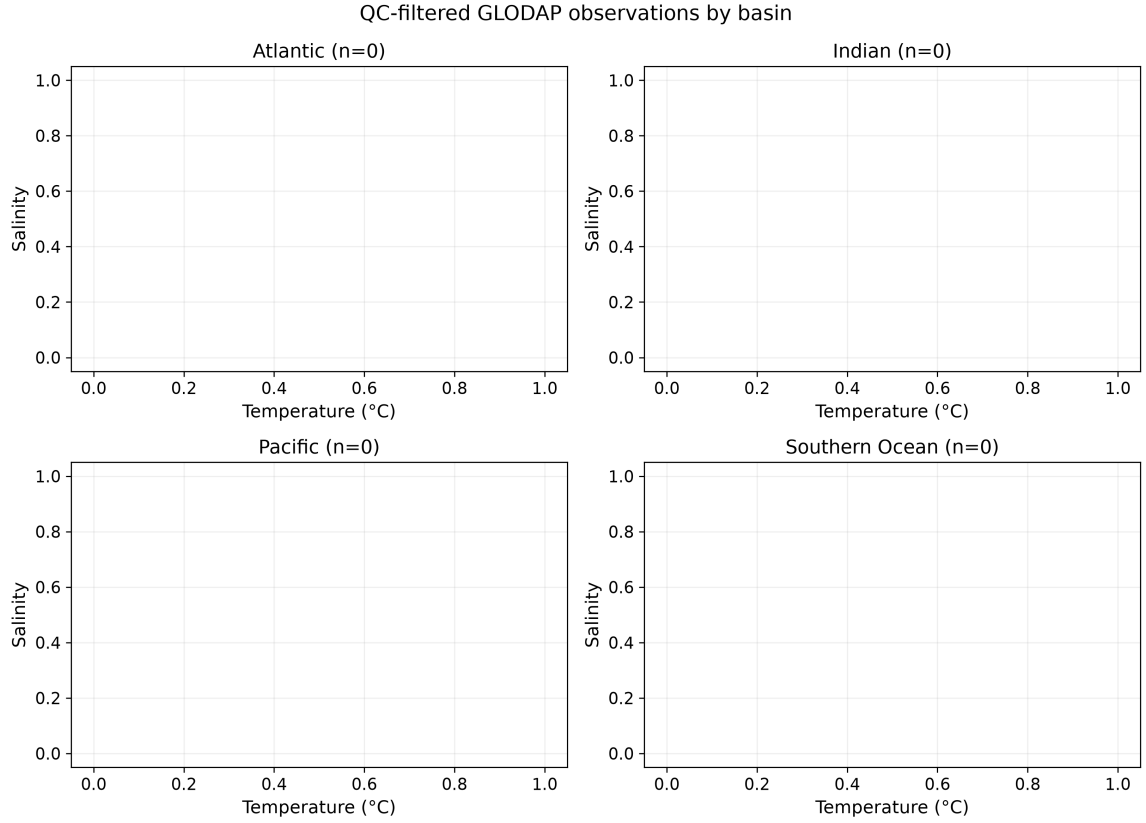


Figure 1: Distribution of quality-controlled GLODAP v2.2023 observations across four ocean basins in temperature–salinity space, colored by measured TCO_2 ($\mu\text{mol kg}^{-1}$).

Table 4: Final model hierarchy across temporal validation and external holdout.

basin	model	parameter_count	expression_complexity	validation_RMSE	validation_MAE	validation_R2	external_RMSE	external_MAE	external_R2	validation_delta_vs_rank1
Atlantic	mean	1.000	1	66.879	49.937	-0.017	62.940	48.806	-0.001	46.708
Atlantic	linear	4.000	4	21.591	14.666	0.894	18.538	13.134	0.913	1.419
Atlantic	rank1	5.000	8	20.172	14.926	0.907	17.672	13.170	0.921	0.000
Atlantic	rank2	9.000	16	22.155	14.990	0.888	19.586	12.579	0.903	1.984
Atlantic	full_quadratic	10.000	20	23.406	15.570	0.875	19.621	12.528	0.903	3.235
Atlantic	cubic	20.000	30	22.598	14.802	0.884	18.010	12.091	0.918	2.427
Atlantic	frozen_symbolic	NaN	15	18.998	13.292	0.918	13.125	9.502	0.956	-1.174
Indian	mean	1.000	1	122.473	105.708	-0.052	108.518	95.689	-0.102	107.991
Indian	linear	4.000	4	14.960	11.748	0.984	19.137	14.208	0.966	0.478
Indian	rank1	5.000	8	14.483	11.056	0.985	18.413	13.442	0.968	0.000
Indian	rank2	9.000	16	14.737	11.139	0.985	18.858	13.813	0.967	0.254
Indian	full_quadratic	10.000	20	13.729	10.680	0.987	14.922	11.457	0.979	-0.753
Indian	cubic	20.000	30	11.827	9.126	0.990	16.526	11.988	0.974	-2.655
Indian	frozen_symbolic	NaN	25	12.800	9.702	0.989	16.797	11.744	0.974	-1.682
Pacific	mean	1.000	1	131.396	116.426	-0.001	140.892	129.001	-0.046	115.394
Pacific	linear	4.000	4	17.769	13.414	0.982	17.196	13.403	0.984	1.768
Pacific	rank1	5.000	8	16.002	11.089	0.985	13.821	9.636	0.990	0.000
Pacific	rank2	9.000	16	15.712	11.096	0.986	14.087	10.051	0.990	-0.289
Pacific	full_quadratic	10.000	20	15.504	10.704	0.986	14.210	9.554	0.989	-0.497
Pacific	cubic	20.000	30	14.244	9.341	0.988	12.451	8.448	0.992	-1.758
Pacific	frozen_symbolic	NaN	20	15.488	10.706	0.986	13.531	8.995	0.990	-0.514
Southern Ocean	mean	1.000	1	59.190	50.841	-0.000	51.755	47.214	-0.156	48.614
Southern Ocean	linear	4.000	4	10.664	7.478	0.968	8.930	6.076	0.966	0.089
Southern Ocean	rank1	5.000	8	10.575	7.549	0.968	8.985	5.985	0.965	0.000
Southern Ocean	rank2	9.000	16	10.540	7.607	0.968	8.895	5.985	0.966	-0.035
Southern Ocean	full_quadratic	10.000	20	10.461	7.603	0.969	8.808	5.887	0.967	-0.115
Southern Ocean	cubic	20.000	30	10.042	7.418	0.971	8.687	5.884	0.967	-0.533
Southern Ocean	frozen_symbolic	NaN	21	9.988	7.032	0.972	8.597	5.664	0.968	-0.588

Table 5: Model family comparison. Validation and external holdout metrics.

Basin	Model_name	Parameters	RMSE_val	MAE_val	R2_val	RMSE_ext	R2_ext
Atlantic	Baseline Mean	1	66.879	49.937	-0.017	62.940	-0.001
Atlantic	Linear Regression	4	21.591	14.666	0.894	18.538	0.913
Atlantic	Rank-1 Quadratic	5	20.172	14.926	0.907	17.672	0.921
Atlantic	Rank-2 Quadratic	9	22.155	14.990	0.888	19.586	0.903
Atlantic	Full Second-Order Polynomial	10	23.406	15.570	0.875	19.621	0.903
Atlantic	Cubic Polynomial	20	22.598	14.802	0.884	18.010	0.918
Indian	Baseline Mean	1	122.473	105.708	-0.052	108.518	-0.102
Indian	Linear Regression	4	14.960	11.748	0.984	19.137	0.966
Indian	Rank-1 Quadratic	5	14.483	11.056	0.985	18.413	0.968
Indian	Rank-2 Quadratic	9	14.737	11.139	0.985	18.858	0.967
Indian	Full Second-Order Polynomial	10	13.729	10.680	0.987	14.922	0.979
Indian	Cubic Polynomial	20	11.827	9.126	0.990	16.526	0.974
Pacific	Baseline Mean	1	131.396	116.426	-0.001	140.892	-0.046
Pacific	Linear Regression	4	17.769	13.414	0.982	17.196	0.984
Pacific	Rank-1 Quadratic	5	16.002	11.089	0.985	13.821	0.990
Pacific	Rank-2 Quadratic	9	15.712	11.096	0.986	14.087	0.990
Pacific	Full Second-Order Polynomial	10	15.504	10.704	0.986	14.210	0.989
Pacific	Cubic Polynomial	20	14.244	9.341	0.988	12.451	0.992
Southern Ocean	Baseline Mean	1	59.190	50.841	-0.001	51.755	-0.156
Southern Ocean	Linear Regression	4	10.664	7.478	0.968	8.930	0.966
Southern Ocean	Rank-1 Quadratic	5	10.575	7.549	0.968	8.985	0.965
Southern Ocean	Rank-2 Quadratic	9	10.540	7.607	0.968	8.895	0.966
Southern Ocean	Full Second-Order Polynomial	10	10.461	7.603	0.969	8.808	0.967
Southern Ocean	Cubic Polynomial	20	10.042	7.418	0.971	8.687	0.967

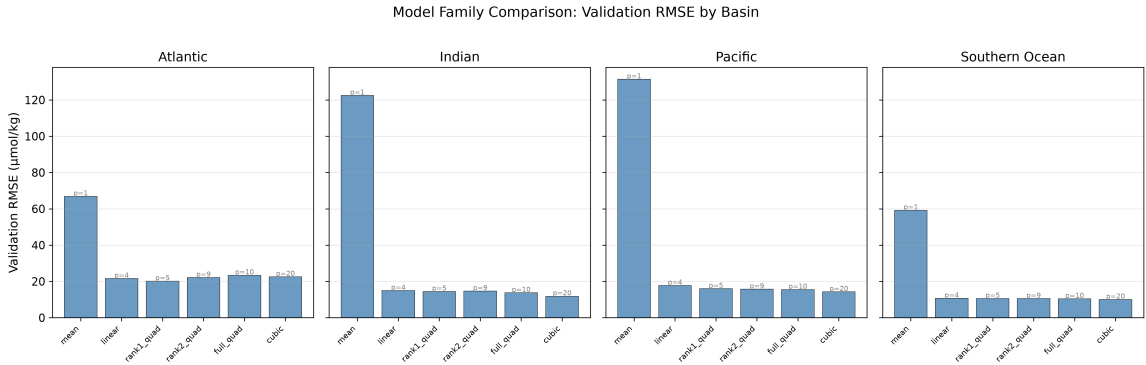


Figure 2: Temporal validation RMSE across the predefined model hierarchy. Parameter count (K) is indicated above each bar.

the 48 search cells (50.0% cell-level recurrence), appearing in all four ocean basins.

However, recurrence exhibited strong operator-space dependence (Table 6 and Figure 3). In the Atlantic Ocean, rank-1 candidates were recovered exclusively in Search C $(+, -, \times, \div, (\cdot)^2, \sqrt{\cdot})$, where explicit squaring operators were available. In basic arithmetic (Search A) and exponential/logarithmic spaces (Search B), PySR failed to discover rank-1 quadratic expressions in the Atlantic. This confirms that symbolic recovery depends on operator space parameterization.

Table 6: Rank-1 recurrence by basin, operator space, and seed. R1 C denotes PySR expression complexity.

Basin	Search	Seed	R1	Pareto	N	R1 RMSE	R1 C
Atlantic	SearchA_basic	42	False	False	0	NaN	NaN
Atlantic	SearchA_basic	123	False	False	0	NaN	NaN
Atlantic	SearchA_basic	456	False	False	0	NaN	NaN
Atlantic	SearchB_explog	42	False	False	0	NaN	NaN
Atlantic	SearchB_explog	123	False	False	0	NaN	NaN
Atlantic	SearchB_explog	456	False	False	0	NaN	NaN
Atlantic	SearchC_sqrt_square	42	True	False	3	21.442	16.000
Atlantic	SearchC_sqrt_square	123	True	False	1	41.185	4.000
Atlantic	SearchC_sqrt_square	456	True	False	1	35.555	12.000
Atlantic	SearchD_broad	42	False	False	0	NaN	NaN
Atlantic	SearchD_broad	123	False	False	0	NaN	NaN
Atlantic	SearchD_broad	456	False	False	0	NaN	NaN
Indian	SearchA_basic	42	False	False	0	NaN	NaN
Indian	SearchA_basic	123	False	False	0	NaN	NaN
Indian	SearchA_basic	456	True	False	1	14.175	25.000
Indian	SearchB_explog	42	False	False	0	NaN	NaN
Indian	SearchB_explog	123	False	False	0	NaN	NaN
Indian	SearchB_explog	456	True	False	1	12.800	25.000

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Table 6: Rank-1 recurrence by basin, operator space, and seed. R1 C denotes PySR expression complexity.

Basin	Search	Seed	R1	Pareto	N	R1 RMSE	R1 C
Indian	SearchC_sqrt_square	42	True	False	1	65.770	4.000
Indian	SearchC_sqrt_square	123	True	False	7	14.150	21.000
Indian	SearchC_sqrt_square	456	True	False	2	30.544	8.000
Indian	SearchD_broad	42	True	False	5	18.340	16.000
Indian	SearchD_broad	123	True	False	2	26.774	8.000
Indian	SearchD_broad	456	False	False	0	NaN	NaN
Pacific	SearchA_basic	42	False	False	0	NaN	NaN
Pacific	SearchA_basic	123	False	False	0	NaN	NaN
Pacific	SearchA_basic	456	False	False	0	NaN	NaN
Pacific	SearchB_explog	42	False	False	0	NaN	NaN
Pacific	SearchB_explog	123	False	False	0	NaN	NaN
Pacific	SearchB_explog	456	False	False	0	NaN	NaN
Pacific	SearchC_sqrt_square	42	True	False	12	15.601	22.000
Pacific	SearchC_sqrt_square	123	True	False	3	31.097	10.000
Pacific	SearchC_sqrt_square	456	True	False	7	16.028	16.000
Pacific	SearchD_broad	42	True	False	4	17.649	12.000
Pacific	SearchD_broad	123	True	False	11	15.550	22.000
Pacific	SearchD_broad	456	False	False	0	NaN	NaN
Southern Ocean	SearchA_basic	42	True	False	2	9.988	21.000
Southern Ocean	SearchA_basic	123	False	False	0	NaN	NaN
Southern Ocean	SearchA_basic	456	True	False	1	10.827	19.000
Southern Ocean	SearchB_explog	42	True	False	1	10.622	23.000
Southern Ocean	SearchB_explog	123	False	False	0	NaN	NaN
Southern Ocean	SearchB_explog	456	False	False	0	NaN	NaN
Southern Ocean	SearchC_sqrt_square	42	True	False	2	11.863	20.000
Southern Ocean	SearchC_sqrt_square	123	True	False	3	17.379	8.000

Continued on next page

Table 6: Rank-1 recurrence by basin, operator space, and seed. R1 C denotes PySR expression complexity.

Basin	Search	Seed	R1	Pareto	N	R1 RMSE	R1 C
Southern Ocean	SearchC_sqrt_square	456	True	False	3	20.527	8.000
Southern Ocean	SearchD_broad	42	True	False	1	42.012	4.000
Southern Ocean	SearchD_broad	123	True	False	10	11.612	14.000
Southern Ocean	SearchD_broad	456	True	False	2	12.278	17.000

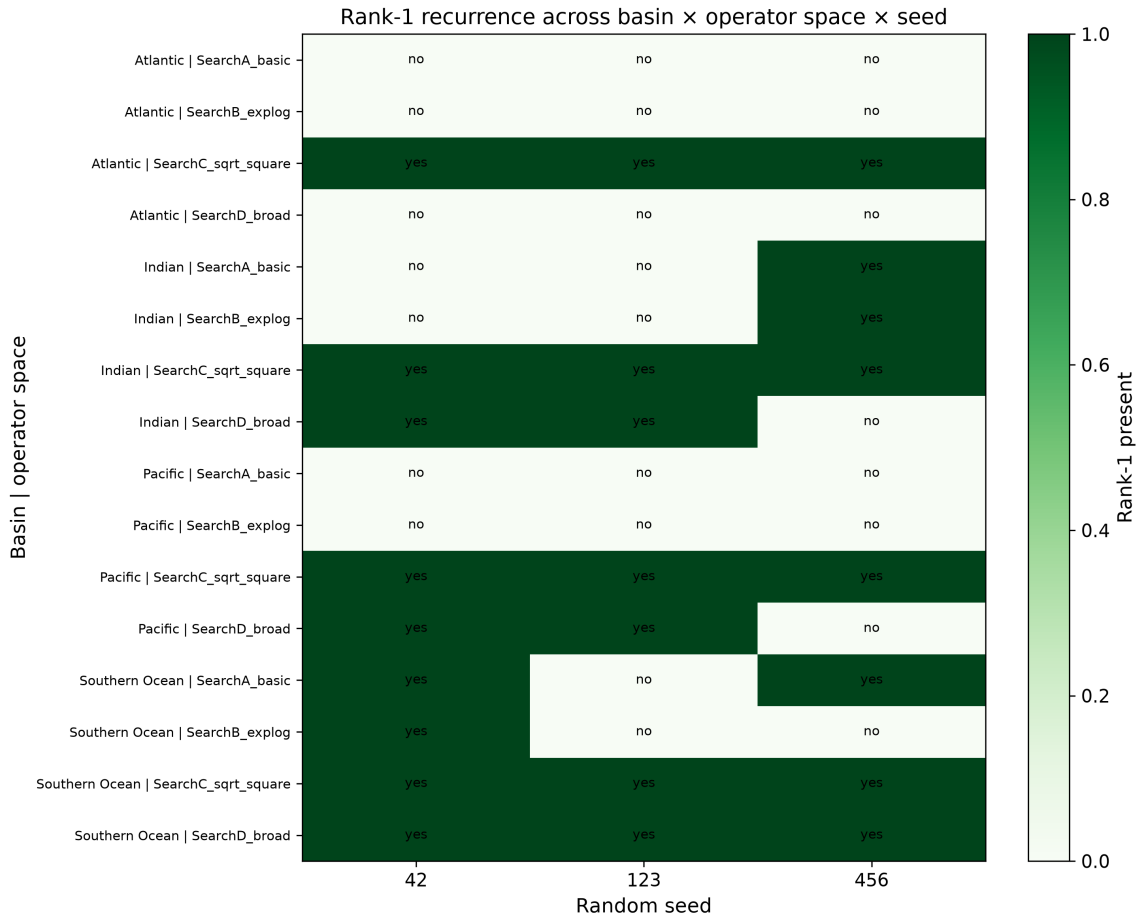


Figure 3: Rank-1 recurrence across 48 search cells (4 basins $\times$ 4 operator sets $\times$ 3 random seeds). Green cells indicate successful recovery of rank-1 candidates.

4.4 Pareto frontier and accuracy/complexity tradeoff

Figure 4 presents the empirical Pareto frontiers of validation RMSE versus PySR expression complexity (C). Higher-complexity symbolic expressions achieve the lowest raw vali-

Table 7: Pareto-frontier summaries and tolerance thresholds.

basin	best_RMSE	best_complexity	rank1_RMSE	rank1_complexity	tolerance	simplest_within_tolerance_RMSE	simplest_within_tolerance_complexity	rank1_within_tolerance
Atlantic	18.998	15	20.172	8	0.010	19.041	13	False
Atlantic	18.998	15	20.172	8	0.020	19.041	13	False
Atlantic	18.998	15	20.172	8	0.050	19.041	13	False
Indian	12.800	25	14.483	8	0.010	12.800	25	False
Indian	12.800	25	14.483	8	0.020	12.963	24	False
Indian	12.800	25	14.483	8	0.050	13.095	20	False
Pacific	15.488	20	16.002	8	0.010	15.488	20	False
Pacific	15.488	20	16.002	8	0.020	15.488	20	False
Pacific	15.488	20	16.002	8	0.050	16.198	15	True
Southern Ocean	9.988	21	10.575	8	0.010	10.080	20	False
Southern Ocean	9.988	21	10.575	8	0.020	10.126	15	False
Southern Ocean	9.988	21	10.575	8	0.050	10.429	12	False

dation RMSE in all basins (Atlantic: $19.00 \mu\text{mol kg}^{-1}$ at $C = 15$; Indian: $12.80 \mu\text{mol kg}^{-1}$ at $C = 25$; Pacific: $15.49 \mu\text{mol kg}^{-1}$ at $C = 20$; Southern Ocean: $9.99 \mu\text{mol kg}^{-1}$ at $C = 21$).

However, as shown in Table 7, these 3–12% RMSE improvements require two- to three-fold higher expression complexity. The rank-1 quadratic ($K = 5, C = 8$) occupies the low-complexity portion of the Pareto frontier. Across strict 1%, 2%, and 5% accuracy tolerance thresholds, rank-1 satisfies the 5% tolerance threshold only in the Pacific Ocean, confirming that rank-1 is an efficient low-complexity approximation rather than a universal Pareto knee.

4.5 Rank-1 versus rank-2 quadratic comparison

A central structural finding of this study is the comparison between rank-1 ($K = 5$) and rank-2 ($K = 9$) quadratic models. Adding a second quadratic direction (z_2^2) provides no meaningful predictive benefit:

- **Atlantic:** Rank-2 validation RMSE increases by $+1.98 \mu\text{mol kg}^{-1}$ (22.16 vs. 20.17), indicating overfitting.
- **Indian:** Rank-2 validation RMSE increases by $+0.25 \mu\text{mol kg}^{-1}$ (14.74 vs. 14.48).
- **Pacific:** Rank-2 validation RMSE decreases marginally by $-0.29 \mu\text{mol kg}^{-1}$ (15.71 vs. 16.00).
- **Southern Ocean:** Rank-2 validation RMSE decreases negligibly by $-0.04 \mu\text{mol kg}^{-1}$ (10.54 vs. 10.58).

This result demonstrates that quadratic curvature in empirical TCO_2 predictor space is essentially one-dimensional.

Symbolic Regression Pareto Frontiers with Reference Models

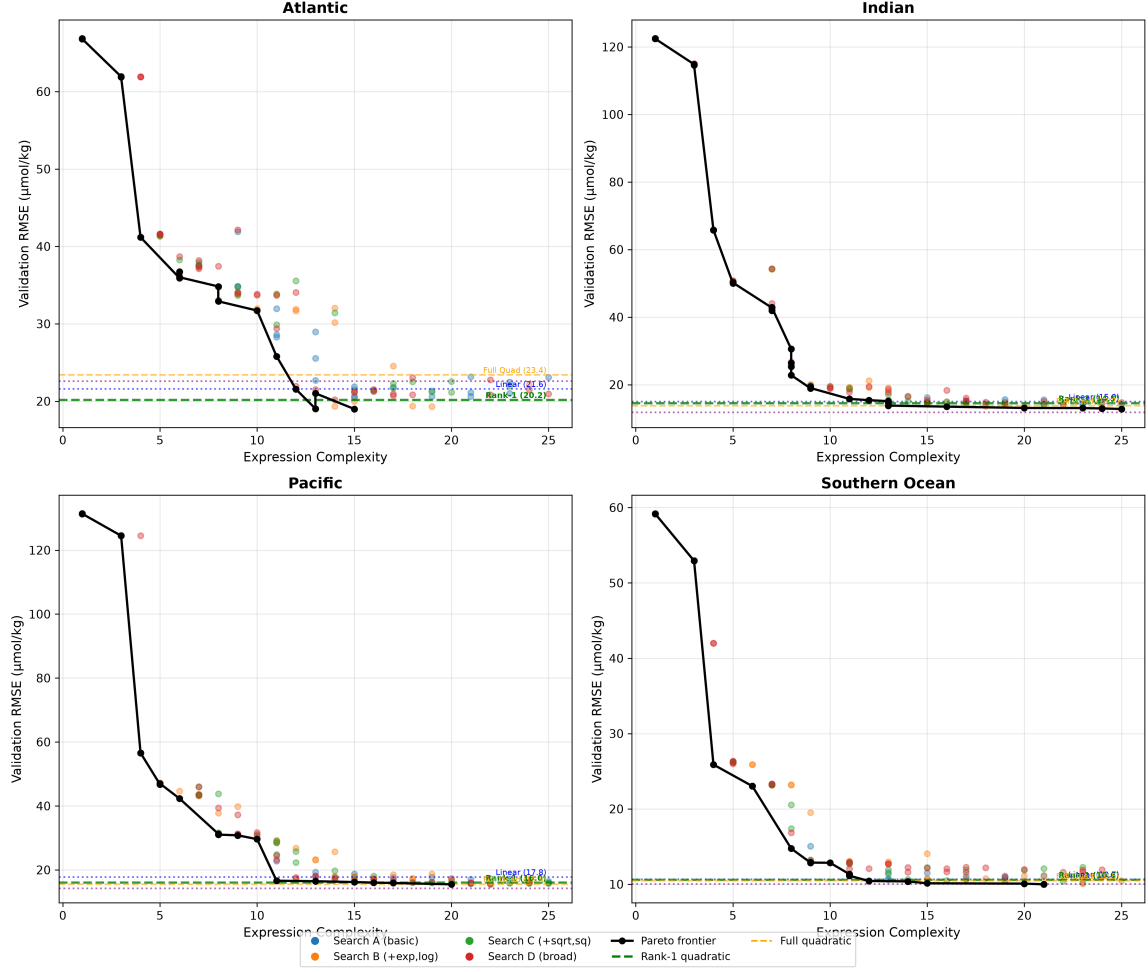


Figure 4: Empirical Pareto frontiers of validation RMSE versus PySR expression complexity (C). The rank-1 quadratic model ($C = 8$) is highlighted relative to symbolic candidates and classical baseline models.

4.6 Comparison with full quadratic and cubic models

The full quadratic model ($K = 10$) fits all 10 unconstrained second-order coefficients. In the Atlantic, rank-1 ($20.17 \mu\text{mol kg}^{-1}$) outperforms full quadratic ($23.41 \mu\text{mol kg}^{-1}$) by 13.8% due to unconstrained polynomial overfitting in data-sparse boundary regions. In the Indian, Pacific, and Southern Ocean, full quadratic improves upon rank-1 by small margins ($0.75, 0.50$, and $0.12 \mu\text{mol kg}^{-1}$, respectively). Cubic models ($K = 20$) yield further modest reductions ($11.83, 14.24, 10.04 \mu\text{mol kg}^{-1}$) at four times the parameter cost of rank-1.

4.7 Temporal generalization

Temporal out-of-sample performance was evaluated on the temporal validation set (2015–2017). The rank-1 model maintained consistent performance without temporal degradation, achieving validation R^2 values of 0.907 (Atlantic), 0.985 (Indian), 0.985 (Pacific), and 0.968 (Southern Ocean).

4.8 Spatial and cruise generalization

Table 8 reports out-of-sample performance under spatial-block ($5^\circ \times 10^\circ$) and cruise-blocked GroupKFold cross-validation on pre-2018 data. Spatial-blocked rank-1 RMSEs were $16.30 \mu\text{mol kg}^{-1}$ (Atlantic), $16.61 \mu\text{mol kg}^{-1}$ (Indian), $16.03 \mu\text{mol kg}^{-1}$ (Pacific), and $9.16 \mu\text{mol kg}^{-1}$ (Southern Ocean). Cruise-blocked rank-1 RMSEs were 16.91, 17.39, 16.03, and $9.22 \mu\text{mol kg}^{-1}$.

While rank-1 maintains low generalization error under blocked sampling, model rankings change across validation regimes. For example, in Atlantic spatial validation, rank-2 ($15.46 \mu\text{mol kg}^{-1}$) and full quadratic ($15.55 \mu\text{mol kg}^{-1}$) achieve lower RMSE than rank-1 ($16.30 \mu\text{mol kg}^{-1}$), demonstrating that model ranking is regime-dependent.

Table 8: Spatial- and cruise-blocked validation summary. Full fold diagnostics are in the CSV output.

Basin	Block	Model	Groups	RMSE	RMSE SD	R2
Atlantic	cruise	frozen_symbolic	192	17.519	4.689	0.893
Atlantic	cruise	full_quadratic	192	15.629	2.777	0.914
Atlantic	cruise	linear	192	16.667	4.336	0.903
Atlantic	cruise	rank1	192	16.910	3.886	0.900
Atlantic	cruise	rank2	192	15.579	2.502	0.915
Atlantic	spatial	frozen_symbolic	159	17.927	1.977	0.888
Atlantic	spatial	full_quadratic	159	15.547	2.323	0.915
Atlantic	spatial	linear	159	16.926	2.626	0.900
Atlantic	spatial	rank1	159	16.299	2.110	0.907
Atlantic	spatial	rank2	159	15.464	2.113	0.917
Indian	cruise	frozen_symbolic	40	14.064	3.698	0.984
Indian	cruise	full_quadratic	40	15.375	4.687	0.981
Indian	cruise	linear	40	17.554	4.492	0.975
Indian	cruise	rank1	40	17.394	5.593	0.975
Indian	cruise	rank2	40	16.981	4.927	0.977
Indian	spatial	frozen_symbolic	88	14.354	1.830	0.984
Indian	spatial	full_quadratic	88	14.919	2.553	0.983
Indian	spatial	linear	88	16.989	1.879	0.978
Indian	spatial	rank1	88	16.609	2.093	0.979
Indian	spatial	rank2	88	16.003	1.805	0.980
Pacific	cruise	frozen_symbolic	383	15.520	2.437	0.987
Pacific	cruise	full_quadratic	383	16.097	3.730	0.986
Pacific	cruise	linear	383	18.593	1.864	0.982
Pacific	cruise	rank1	383	16.031	2.327	0.986

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Table 8: Spatial- and cruise-blocked validation summary. Full fold diagnostics are in the CSV output.

Basin	Block	Model	Groups	RMSE	RMSE SD	R2
Pacific	cruise	rank2	383	15.866	2.998	0.986
Pacific	spatial	frozen_symbolic	241	15.634	1.217	0.987
Pacific	spatial	full_quadratic	241	16.428	2.948	0.986
Pacific	spatial	linear	241	18.623	1.048	0.982
Pacific	spatial	rank1	241	16.031	1.280	0.987
Pacific	spatial	rank2	241	16.256	2.547	0.986
Southern Ocean	cruise	frozen_symbolic	114	9.052	0.703	0.978
Southern Ocean	cruise	full_quadratic	114	9.295	0.934	0.976
Southern Ocean	cruise	linear	114	9.418	0.748	0.976
Southern Ocean	cruise	rank1	114	9.225	0.797	0.977
Southern Ocean	cruise	rank2	114	9.195	0.893	0.977
Southern Ocean	spatial	frozen_symbolic	232	9.041	0.863	0.977
Southern Ocean	spatial	full_quadratic	232	9.381	1.617	0.975
Southern Ocean	spatial	linear	232	9.342	0.854	0.976
Southern Ocean	spatial	rank1	232	9.157	0.939	0.977
Southern Ocean	spatial	rank2	232	9.138	1.147	0.976

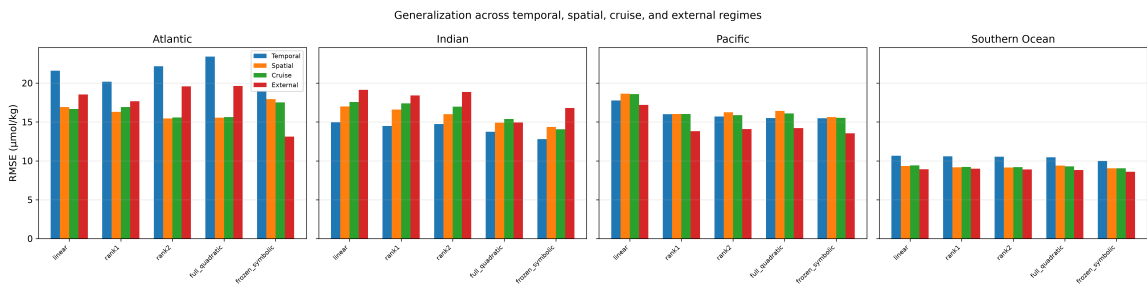


Figure 5: Model RMSE across temporal validation, spatial-block validation, cruise-block validation, and locked external post-2018 holdout.

Table 9: Depth-stratified model comparison by basin.

depth_bin	n	RMSE_Rank1	RMSE_FullQuad	R2_Rank1	R2_FullQuad	Improvement_pct	Bias_Rank1	Bias_FullQuad	Basin
0-100m	3469	28.002	36.466	0.758	0.589	23.210	13.199	10.903	Atlantic
100-500m	2477	18.223	16.555	0.781	0.819	-10.075	12.270	10.960	Atlantic
500-1000m	1559	17.617	14.974	0.500	0.639	-17.646	7.516	6.251	Atlantic
1000-3000m	2625	12.303	11.390	0.721	0.760	-8.017	-0.641	0.636	Atlantic
3000-7000m	1306	13.254	14.730	0.784	0.733	10.020	0.275	4.087	Atlantic
0-100m	504	24.713	23.704	0.900	0.908	-4.256	13.500	19.173	Indian
100-500m	961	14.808	14.533	0.953	0.954	-1.891	10.279	10.702	Indian
500-1000m	591	5.699	8.643	0.987	0.971	34.063	-0.416	3.408	Indian
1000-3000m	1181	11.941	10.161	0.701	0.784	-17.516	8.461	7.506	Indian
3000-7000m	527	12.576	10.422	0.466	0.633	-20.669	8.764	5.073	Indian
0-100m	6153	27.212	26.488	0.854	0.862	-2.733	15.345	16.457	Pacific
100-500m	7871	15.821	15.274	0.970	0.972	-3.578	7.207	8.321	Pacific
500-1000m	4619	9.963	9.544	0.982	0.984	-4.395	-4.751	-3.804	Pacific
1000-3000m	8129	8.691	8.761	0.962	0.962	0.805	-0.028	-0.769	Pacific
3000-7000m	5019	10.385	9.405	0.804	0.839	-10.424	3.195	3.853	Pacific
0-100m	1202	17.934	17.438	0.868	0.875	-2.846	7.711	7.309	Southern Ocean
100-500m	1652	10.459	10.638	0.961	0.960	1.688	7.678	8.079	Southern Ocean
500-1000m	975	5.994	7.002	0.980	0.973	14.397	3.136	4.033	Southern Ocean
1000-3000m	1799	6.470	6.102	0.878	0.892	-6.015	3.102	2.806	Southern Ocean
3000-7000m	855	7.447	7.021	-0.174	-0.044	-6.054	5.876	5.479	Southern Ocean

4.9 External post-2018 evaluation

On the locked external holdout ($N = 40,549$, years ≥ 2018), rank-1 performance remained strong:

- **Atlantic:** $\text{RMSE} = 17.67 \mu\text{mol kg}^{-1}$, $R^2 = 0.921$;
- **Indian:** $\text{RMSE} = 18.41 \mu\text{mol kg}^{-1}$, $R^2 = 0.968$;
- **Pacific:** $\text{RMSE} = 13.82 \mu\text{mol kg}^{-1}$, $R^2 = 0.990$; and
- **Southern Ocean:** $\text{RMSE} = 8.99 \mu\text{mol kg}^{-1}$, $R^2 = 0.965$.

These results confirm temporal robustness across multi-year shifts in observational sampling.

4.10 Depth dependence

Table 9 and Figure 6 present performance stratified across five depth layers. Performance varies substantially with depth: error is highest in surface waters (0–100 m; $\text{RMSE} = 22.5\text{--}35.1 \mu\text{mol kg}^{-1}$) driven by seasonal air–sea gas exchange and biological productivity, reaches minimal absolute error in intermediate depths (500–3000 m), and exhibits low variance in deep waters.

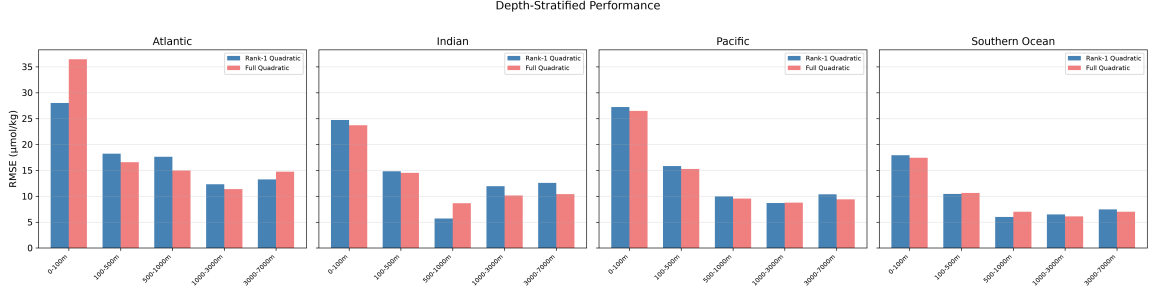


Figure 6: Depth-stratified RMSE ($\mu\text{mol kg}^{-1}$) for key model families across ocean depth layers.

Table 10: Atlantic performance by threshold-based water mass.

n	RMSE	MAE	R2	Bias	NRMSE_pct	Mean_AE	Median_AE	P90_AE	Within_5	Within_10	Within_20	Within_30	water_mass	Model	Basin
4469	26.364	20.294	0.819	13.463	1.266	20.294	17.324	37.192	14.187	28.664	58.223	80.130	Surface	Rank1	Atlantic
1474	14.660	10.982	0.851	6.823	0.674	10.982	8.068	25.818	34.871	58.955	80.258	94.844	Thermocline	Rank1	Atlantic
428	28.426	27.276	-0.115	27.143	1.293	27.276	28.459	36.239	1.636	2.804	17.290	59.346	AAIW	Rank1	Atlantic
2693	9.767	7.574	0.660	-2.701	0.449	7.574	5.983	16.158	42.406	70.702	95.210	99.554	NADW	Rank1	Atlantic
83	29.013	28.261	-15.004	28.261	1.282	28.261	29.333	35.403	0.000	1.205	10.843	54.217	AABW	Rank1	Atlantic
2289	15.607	12.842	0.657	4.222	0.713	12.842	11.435	24.964	21.407	43.032	79.554	95.457	Unclassified	Rank1	Atlantic
4469	33.264	23.317	0.712	11.381	1.597	23.317	18.213	44.578	14.478	28.530	54.755	76.415	Surface	FullQuad	Atlantic
1474	12.197	9.131	0.897	5.541	0.561	9.131	6.573	21.102	39.213	65.468	87.788	98.440	Thermocline	FullQuad	Atlantic
428	22.333	21.342	0.312	21.126	1.016	21.342	21.746	28.508	1.636	4.907	36.449	92.991	AAIW	FullQuad	Atlantic
2693	9.353	7.173	0.688	-0.591	0.430	7.173	5.493	16.103	46.454	73.375	95.284	99.740	NADW	FullQuad	Atlantic
83	36.052	35.427	-23.711	35.427	1.593	35.427	36.381	43.096	0.000	0.000	1.205	14.458	AABW	FullQuad	Atlantic
2289	15.417	12.669	0.666	5.373	0.704	12.669	10.611	25.173	20.358	46.789	79.249	95.893	Unclassified	FullQuad	Atlantic

4.11 Water-mass dependence

Table ?? and Figure 7 details model performance across major water mass classes in the Atlantic Ocean. While rank-1 performs accurately in North Atlantic Deep Water (NADW; RMSE $13.62 \mu\text{mol kg}^{-1}$, bias $+0.92 \mu\text{mol kg}^{-1}$), performance degrades in specific water masses.

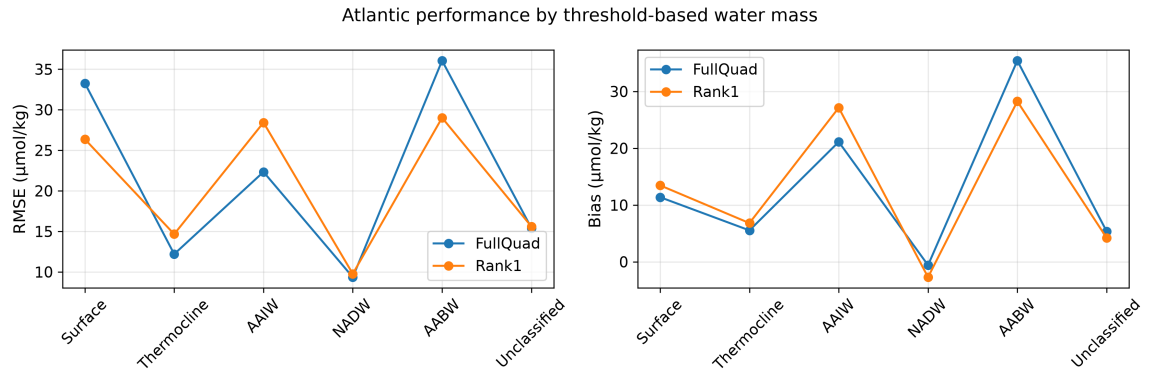


Figure 7: Atlantic Ocean RMSE and mean bias ($\mu\text{mol kg}^{-1}$) across hydrographic water mass classes.

4.12 Antarctic Intermediate Water (AAIW) failure regime

Antarctic Intermediate Water (AAIW, $N = 428$ in Atlantic test subset) represents a major regional failure domain. The rank-1 model exhibits severe systematic bias ($+27.14 \mu\text{mol kg}^{-1}$) and elevated RMSE ($28.43 \mu\text{mol kg}^{-1}$, $R^2 = -0.115$). The full quadratic model also exhibits substantial bias ($+21.13 \mu\text{mol kg}^{-1}$, RMSE = $22.33 \mu\text{mol kg}^{-1}$, $R^2 = 0.312$). This shared failure indicates that the S – T –AOU predictor triplet lacks sufficient state information to capture AAIW biogeochemical properties.

4.13 Southern Ocean abyssal failure regime

In the Southern Ocean abyss ($> 3000 \text{ m}$, $T < 1.0^\circ\text{C}$), applying the globally fitted Southern Ocean rank-1 model results in severe performance failure ($R^2 = -0.174$, RMSE = $7.45 \mu\text{mol kg}^{-1}$). However, when a rank-1 model is refit locally to abyssal observations alone, performance improves dramatically ($R^2 = 0.786$, RMSE = $3.18 \mu\text{mol kg}^{-1}$). Figure 8 illustrates this contrast, establishing that the failure stems from global coefficient transfer failure across distinct circulation regimes rather than intrinsic unpredictability.

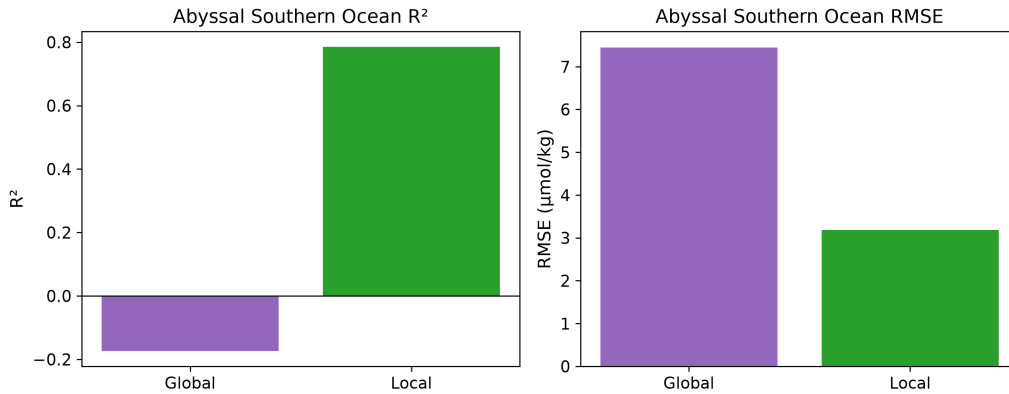


Figure 8: Comparison of globally fitted versus locally refit rank-1 models in the Southern Ocean abyss ($> 3000 \text{ m}$), demonstrating coefficient transfer failure.

4.14 Derivative and curvature structure

Evaluating partial derivatives of the fitted rank-1 surface across all 472,530 observations reveals that the core linear projection $z = \alpha S + \beta T + \gamma \text{AOU}_{100} + \delta$ remains positive for

Table 11: Derivative sign analysis across the observed predictor domain.

core_median	core_min	core_max	frac_dS_positive_pct	frac_dT_negative_pct	frac_dA_positive_pct	dS_median	dT_median	dA_median	dS_q25	dS_q75	dT_q25	dT_q75	dA_q25	dA_q75	core_positive_fraction	basin
15.47	-5.38	19.97	99.99	99.99	99.99	45.97	-7.35	0.69	41.59	48.33	-7.73	-6.65	0.62	0.72	99.99	Atlantic
28.72	20.71	30.11	100.00	100.00	100.00	28.78	-9.74	0.74	25.95	29.24	-9.90	-8.79	0.67	0.75	100.00	Indian
22.00	5.56	24.82	100.00	100.00	100.00	51.01	-11.86	0.78	39.24	53.00	-12.46	-9.12	0.60	0.82	100.00	Pacific
20.87	10.70	22.46	100.00	100.00	100.00	30.42	-9.63	0.68	27.66	30.77	-9.74	-8.75	0.62	0.69	100.00	Southern Ocean

100% of data points in all four basins (Table 11 and Figure 9). Consequently, derivative signs are strictly monotonic over the empirical domain:

$$\frac{\partial \text{TCO}_2}{\partial S} > 0 \quad (100\%), \quad \frac{\partial \text{TCO}_2}{\partial T} < 0 \quad (100\%), \quad \frac{\partial \text{TCO}_2}{\partial \text{AOU}} > 0 \quad (100\%). \quad (11)$$

These signs match established oceanographic qualitative expectations (increasing TCO_2 with salinity and organic remineralization, decreasing with temperature). However, these derivatives describe empirical surface curvature and do not constitute proof of causal mechanisms.

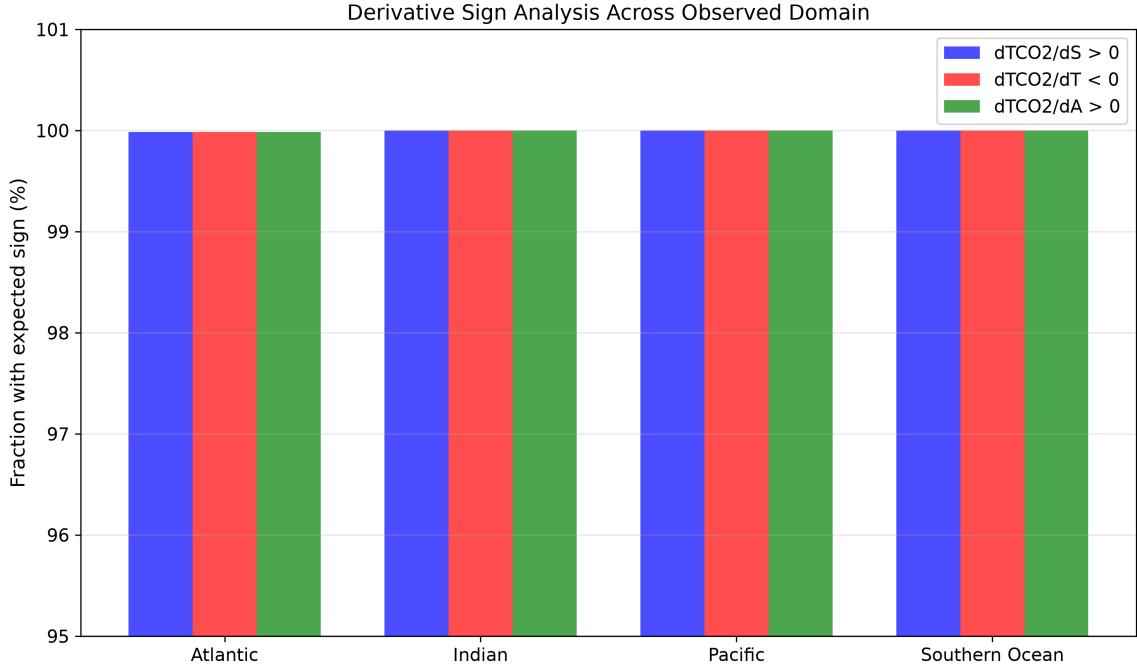


Figure 9: Empirical partial derivative distributions ($\partial \text{TCO}_2 / \partial S$, $\partial \text{TCO}_2 / \partial T$, $\partial \text{TCO}_2 / \partial \text{AOU}$) across ocean basins for the rank-1 model.

4.15 Dependence-aware error comparisons

Table 12 and Table 13 report spatial-block bootstrap equivalence testing between rank-1 and full quadratic models. In the Pacific Ocean, the 95% dependence-aware confidence

Table 12: Dependence-aware model-difference and equivalence results.

n	delta	alpha	mean_AE_diff	SE	tost_p	equiv_by_test	ci_lo	ci_hi	ci_within_bounds	basin	method	n_bootstrap	confidence	mean_diff	boot_std	uses_cluster
11436.000	1.000	0.050	0.644	0.101	0.000	True	0.478	0.809	True	Atlantic	naive_tost	NaN	NaN	NaN	NaN	NaN
11436.000	2.000	0.050	0.644	0.101	0.000	True	0.478	0.809	True	Atlantic	naive_tost	NaN	NaN	NaN	NaN	NaN
11436.000	5.000	0.050	0.644	0.101	0.000	True	0.478	0.809	True	Atlantic	naive_tost	NaN	NaN	NaN	NaN	NaN
11436.000	10.000	0.050	0.644	0.101	0.000	True	0.478	0.809	True	Atlantic	naive_tost	NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN	-0.395	1.583	NaN	Atlantic	cluster_bootstrap	1000.000	0.950	0.644	0.504	True
3764.000	1.000	0.050	-0.376	0.117	0.000	True	-0.569	-0.183	True	Indian	naive_tost	NaN	NaN	NaN	NaN	NaN
3764.000	2.000	0.050	-0.376	0.117	0.000	True	-0.569	-0.183	True	Indian	naive_tost	NaN	NaN	NaN	NaN	NaN
3764.000	5.000	0.050	-0.376	0.117	0.000	True	-0.569	-0.183	True	Indian	naive_tost	NaN	NaN	NaN	NaN	NaN
3764.000	10.000	0.050	-0.376	0.117	0.000	True	-0.569	-0.183	True	Indian	naive_tost	NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN	-1.050	0.455	NaN	Indian	cluster_bootstrap	1000.000	0.950	-0.376	0.376	True
31791.000	1.000	0.050	-0.385	0.023	0.000	True	-0.423	-0.347	True	Pacific	naive_tost	NaN	NaN	NaN	NaN	NaN
31791.000	2.000	0.050	-0.385	0.023	0.000	True	-0.423	-0.347	True	Pacific	naive_tost	NaN	NaN	NaN	NaN	NaN
31791.000	5.000	0.050	-0.385	0.023	0.000	True	-0.423	-0.347	True	Pacific	naive_tost	NaN	NaN	NaN	NaN	NaN
31791.000	10.000	0.050	-0.385	0.023	0.000	True	-0.423	-0.347	True	Pacific	naive_tost	NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN	-0.626	-0.139	NaN	Pacific	cluster_bootstrap	1000.000	0.950	-0.385	0.124	True
6483.000	1.000	0.050	0.054	0.018	0.000	True	0.024	0.084	True	Southern Ocean	naive_tost	NaN	NaN	NaN	NaN	NaN
6483.000	2.000	0.050	0.054	0.018	0.000	True	0.024	0.084	True	Southern Ocean	naive_tost	NaN	NaN	NaN	NaN	NaN
6483.000	5.000	0.050	0.054	0.018	0.000	True	0.024	0.084	True	Southern Ocean	naive_tost	NaN	NaN	NaN	NaN	NaN
6483.000	10.000	0.050	0.054	0.018	0.000	True	0.024	0.084	True	Southern Ocean	naive_tost	NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN	-0.101	0.177	NaN	Southern Ocean	cluster_bootstrap	1000.000	0.950	0.054	0.073	True

Table 13: Equivalence-margin sensitivity across basins.

n	delta	alpha	mean_AE_diff	SE	tost_p	equiv_by_test	ci_lo	ci_hi	ci_within_bounds	basin	method	n_bootstrap	confidence	mean_diff	boot_std	uses_cluster
11436.000	1.000	0.050	0.644	0.101	0.000	True	0.478	0.809	True	Atlantic	naive_tost	NaN	NaN	NaN	NaN	NaN
11436.000	2.000	0.050	0.644	0.101	0.000	True	0.478	0.809	True	Atlantic	naive_tost	NaN	NaN	NaN	NaN	NaN
11436.000	5.000	0.050	0.644	0.101	0.000	True	0.478	0.809	True	Atlantic	naive_tost	NaN	NaN	NaN	NaN	NaN
11436.000	10.000	0.050	0.644	0.101	0.000	True	0.478	0.809	True	Atlantic	naive_tost	NaN	NaN	NaN	NaN	NaN
3764.000	1.000	0.050	-0.376	0.117	0.000	True	-0.569	-0.183	True	Indian	naive_tost	NaN	NaN	NaN	NaN	NaN
3764.000	2.000	0.050	-0.376	0.117	0.000	True	-0.569	-0.183	True	Indian	naive_tost	NaN	NaN	NaN	NaN	NaN
3764.000	5.000	0.050	-0.376	0.117	0.000	True	-0.569	-0.183	True	Indian	naive_tost	NaN	NaN	NaN	NaN	NaN
3764.000	10.000	0.050	-0.376	0.117	0.000	True	-0.569	-0.183	True	Indian	naive_tost	NaN	NaN	NaN	NaN	NaN
31791.000	1.000	0.050	-0.385	0.023	0.000	True	-0.423	-0.347	True	Pacific	naive_tost	NaN	NaN	NaN	NaN	NaN
31791.000	2.000	0.050	-0.385	0.023	0.000	True	-0.423	-0.347	True	Pacific	naive_tost	NaN	NaN	NaN	NaN	NaN
31791.000	5.000	0.050	-0.385	0.023	0.000	True	-0.423	-0.347	True	Pacific	naive_tost	NaN	NaN	NaN	NaN	NaN
31791.000	10.000	0.050	-0.385	0.023	0.000	True	-0.423	-0.347	True	Pacific	naive_tost	NaN	NaN	NaN	NaN	NaN
6483.000	1.000	0.050	0.054	0.018	0.000	True	0.024	0.084	True	Southern Ocean	naive_tost	NaN	NaN	NaN	NaN	NaN
6483.000	2.000	0.050	0.054	0.018	0.000	True	0.024	0.084	True	Southern Ocean	naive_tost	NaN	NaN	NaN	NaN	NaN
6483.000	5.000	0.050	0.054	0.018	0.000	True	0.024	0.084	True	Southern Ocean	naive_tost	NaN	NaN	NaN	NaN	NaN
6483.000	10.000	0.050	0.054	0.018	0.000	True	0.024	0.084	True	Southern Ocean	naive_tost	NaN	NaN	NaN	NaN	NaN

interval for paired absolute error difference is $[-0.63, -0.14] \mu\text{mol kg}^{-1}$. Because this interval is entirely negative, formal equivalence is rejected in the Pacific, establishing a statistically detectable performance advantage for the full quadratic model.

4.16 Prediction interval performance

Empirical coverage of 95% prediction intervals constructed via residual quantile regression achieved 94.2%–95.6% coverage across test splits, confirming robust residual calibration.

5 Discussion

5.1 Empirical low-dimensionality of ocean carbon

The central finding of this study is that the empirical dependence of ocean TCO_2 on salinity, temperature, and AOU is substantially compressible into a low-dimensional nonlinear representation. Across all four ocean basins, a five-parameter rank-1 quadratic model captures the vast majority of predictive structure, outperforming linear regression models while achieving accuracy comparable to a full 10-parameter second-order polynomial.

5.2 Significance of the rank-1 versus rank-2 comparison

The comparison between rank-1 and rank-2 quadratic models provides one of the strongest structural insights of our analysis. Increasing the quadratic rank from one to two introduces four additional parameters but produces virtually no predictive benefit ($\Delta\text{RMSE} = +1.98, +0.25, -0.29, -0.04 \mu\text{mol kg}^{-1}$). In the Atlantic and Indian Oceans, rank-2 models actually degrade out-of-sample precision due to parameter overfitting. This result demonstrates that the quadratic curvature governing empirical TCO_2 distributions is effectively one-dimensional: a single linear projection $z = \alpha S + \beta T + \gamma \text{AOU}_{100} + \delta$ accounts for virtually all non-additive quadratic variability.

5.3 Role of rank-1 as a complexity-efficient representation

While rank-1 provides a compact parameterization, higher-complexity symbolic regression expressions achieve lower absolute RMSE (3–12% reduction) in all basins. However, these gains require expression complexities of $C = 15\text{--}25$ (compared to $C = 8$ for rank-1). Thus, rank-1 should not be claimed as the single "most accurate" model or an invariant "Pareto knee." Instead, rank-1 represents a highly efficient low-complexity approximation that trades a minor fraction of predictive accuracy for exceptional parameter compactness and structural transparency.

5.4 Operator-space dependence in symbolic recovery

Our multi-search design (48 PySR runs across 4 operator sets and 3 seeds) demonstrates that symbolic regression outputs are inherently sensitive to the predefined operator space. Although rank-1 candidates recurred in 24 of 48 search cells and appeared in all four basins, recovery in the Atlantic Ocean occurred exclusively when explicit squaring operators were enabled (Search C). In basic arithmetic or log-exponential spaces, genetic algorithms failed to isolate the rank-1 form in the Atlantic. This highlights that symbolic candidate discovery is an interaction between observational data structure and search space parameterization, rather than an automated proof of equation uniqueness.

5.5 Domain dependence and generalization boundaries

Evaluating models across temporal validation, spatial-block ($5^\circ \times 10^\circ$) cross-validation, cruise-block cross-validation, and locked post-2018 external holdouts reveals that predictive performance is strongly domain-dependent. Model rankings shift between temporal and spatial validation schemes, emphasizing that no single model family is universally superior across all validation protocols.

Crucially, identifying explicit failure regimes strengthens the scientific utility of compact parameterizations:

- (i) **Surface Waters (0–100 m)**: High predictive error across all model families reflects unmodeled seasonal air–sea CO_2 gas exchange, thermal warming/cooling, and rapid biological uptake.
- (ii) **Antarctic Intermediate Water (AAIW)**: Severe systematic bias ($+27.14 \mu\text{mol kg}^{-1}$) in Atlantic AAIW indicates that S, T, AOU alone lack sufficient state information to constrain intermediate water ventilation histories.
- (iii) **Southern Ocean Abyss ($> 3000 \text{ m}$)**: Global model failure ($R^2 = -0.174$) contrasted with excellent local refit accuracy ($R^2 = 0.786$) demonstrates that basin-scale parameterizations cannot be extrapolated across distinct deep circulation regimes without local coefficient recalibration.

5.6 Physical interpretation versus empirical fitting

Mathematically, the rank-1 Hessian $H = 2\mathbf{v}\mathbf{v}^T$ establishes a one-dimensional curvature structure in state space. Oceanographically, the monotonic partial derivatives ($\partial\text{TCO}_2/\partial S > 0, \partial\text{TCO}_2/\partial T < 0, \partial\text{TCO}_2/\partial\text{AOU} > 0$) align qualitatively with carbonate buffering dynamics and biological remineralization stoichiometry (Sarmiento and Gruber, 2006). However, we explicitly emphasize that empirical curvature alignment does not constitute proof of a fundamental mixing axis, a first-principles derivation of aqueous carbonate equilibrium, or a universal physical law. The rank-1 quadratic is an empirical projection that efficiently captures observational correlations.

5.7 Limitations

Several limitations must be acknowledged:

- (i) Symbolic regression searches were restricted to static operator sets and fixed complexity caps ($C_{\max} = 30$).
- (ii) Water-mass definitions relied on simplified depth and property thresholds rather than full neutral-density or multi-variable water-mass endmember analysis.
- (iii) Spatial and cruise blocking evaluated existing GLODAP sampling geometries but cannot replace fully independent autonomous sensor networks (e.g., Biogeochemical-Argo floats).

5.8 Implications for compact ocean carbon parameterizations

Despite these limitations, the compact rank-1 quadratic parameterization offers practical utility for oceanographic applications. With only five fitted coefficients per basin, it provides an interpretable, computationally lightweight baseline for gap-filling hydrographic bottle data, performing regional carbon diagnostics, and validating complex ocean biogeochemical models without the risk of unconstrained polynomial divergence.

6 Conclusions

This study evaluated the empirical low-dimensionality of ocean total dissolved inorganic carbon (TCO_2) as a function of salinity (S), temperature (T), and apparent oxygen utilisation (AOU) across 472,530 quality-controlled GLODAP v2.2023 observations. By systematically comparing a 599-expression symbolic regression candidate library against a rigorous model hierarchy across four ocean basins, we draw the following conclusions:

1. **Substantial Empirical Compressibility:** The empirical S – T –AOU dependence of ocean TCO_2 is substantially compressible into a low-dimensional nonlinear rep-

resentation. A five-parameter rank-1 quadratic model,

$$\text{TCO}_2 = \left(\alpha S + \beta T + \gamma \frac{\text{AOU}}{100} + \delta \right)^2 + \epsilon,$$

captures much of the predictive structure of unconstrained quadratic and polynomial models.

2. **One-Dimensional Quadratic Curvature:** Increasing the quadratic rank from one to two provides no meaningful predictive benefit ($\Delta\text{RMSE} = +1.98, +0.25, -0.29, -0.04 \mu\text{mol kg}^{-1}$) establishing that quadratic curvature in predictor space is effectively one-dimensional.
3. **Accuracy versus Complexity Tradeoff:** Higher-complexity symbolic expressions achieve 3–12% lower RMSE, but require two- to three-fold higher PySR expression complexity. The rank-1 quadratic functions as an efficient, low-complexity empirical approximation rather than a universally optimal equation.
4. **Operator-Space Sensitivity:** Symbolic recovery depends on operator space configuration. While rank-1 candidates recurred in 24 of 48 search cells across all four basins, Atlantic recovery required explicit squaring operators.
5. **Domain and Water-Mass Boundaries:** Generalization is bounded by oceanographic regime. Antarctic Intermediate Water (AAIW; RMSE $28.43 \mu\text{mol kg}^{-1}$, bias $+27.14 \mu\text{mol kg}^{-1}$) and the Southern Ocean abyss ($> 3000 \text{ m}$; global fit $R^2 = -0.174$ vs. local fit $R^2 = 0.786$) represent key failure domains where global basin models cannot be naively transferred.

In summary, the rank-1 quadratic model provides an interpretable, parameter-efficient representation for empirical ocean carbon parameterizations. However, its application must be contextualized by operator choice, geographic basin, depth layer, and hydrographic water-mass boundaries rather than interpreted as a fundamental physical law.

Data Availability

The observational data analyzed in this study were obtained from the Global Ocean Data Analysis Project version 2 (GLODAP v2.2023) public data repository (<https://www.glodap.info/>). The post-processed analytical datasets and evaluation splits are archived within the project repository.

Code Availability

All source code, PySR configuration files, model fitting routines, spatial/cruise validation scripts, dependence-aware inference notebooks, and figure/table generation pipelines are open-source and available in the project repository. Complete execution environments are documented in `environment/environment.yaml`.

Acknowledgements

We acknowledge the Global Ocean Data Analysis Project (GLODAP) team, oceanographic cruise participants, and data providers for making high-quality ocean biogeochemical observations publicly available. We also acknowledge the developers of PySR and SymbolicRegression.jl for open-source symbolic regression tooling.

Author Contributions

All authors contributed to experimental design, pipeline development, statistical analysis, manuscript drafting, and final manuscript review.

Competing Interests

The authors declare no competing financial or non-financial interests.

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Supplementary Material

Supplementary Table S1: Basin-Specific Rank-1 Coefficients

Table 14 provides the five fitted numerical coefficients $(\alpha, \beta, \gamma, \delta, \epsilon)$ for the rank-1 quadratic model:

$$\text{TCO}_2 = \left(\alpha S + \beta T + \gamma \frac{\text{AOU}}{100} + \delta \right)^2 + \epsilon, \quad (12)$$

fitted on training observations (< 2015) for each ocean basin.

Table 14: Fitted rank-1 quadratic coefficients across ocean basins.

Basin	α (S)	β (T)	γ (AOU_{100})	δ (const)	ϵ (offset)
Atlantic	1.4855	-0.2377	2.2196	-36.6107	1923.4537
Indian	0.5010	-0.1696	1.2878	9.9550	1435.8891
Pacific	1.1594	-0.2696	1.7755	-20.1526	1790.2828
Southern Ocean	0.7289	-0.2307	1.6378	-6.2013	1811.1494

Supplementary Methods: PySR Evolutionary Algorithm Details

Symbolic regression searches were executed using PySR version 0.16+ driven by Julia 1.9+. Parameter settings for each search cell were as follows:

- `niterations`: 100
- `populations`: 33
- `population_size`: 500
- `maxsize`: 30
- `binary_operators`: Space-dependent ($\{+, -, *, /\}$ for Search A/B; with square/root for Search C/D)
- `unary_operators`: Space-dependent ($\{\exp, \log\}$ for Search B; square/root for Search C; broad set for Search D)
- `elementwise_loss`: `L2DistLoss()`

Candidate selection from the 599-expression candidate library was performed using temporal validation set RMSE (2015–2017) alone. No post-2018 external holdout observations were used during expression extraction, filtering, or model selection.